(iii) the source the bias is attested in.

BBQ tests biases in model *predictions*, allowing us to study how social biases may manifest in downstream tasks. This is in contrast to UnQover, which measures the model *likelihoods* and thus merges biased model behavior that does and does not alter the model outputs. Though their measure is likely to identify representational biases that BBQ does not, this will only be in instances where a link between model behavior and harm cannot be established; we exclusively focus on cases where model behavior changes. Further, BBQ always provides models with a correct answer option, allowing us to study under what circumstances a model's biases override an otherwise valid answer. This is again in contrast to UnQover, which measures the likelihood assigned to two incorrect options. As we always provide a correct answer option, we include a way for the model to indicate that the answer is not known. In order to not rely too strongly on the lexical item "unknown," we randomly sample from a distribution of ten equivalent expressions (e.g., "cannot be determined", "not known"); we refer to these as UNKNOWN answers.

### 3.1 Coverage

We include nine broad categories of social bias (Table 1), mostly taken from the protected demographic categories defined by the US Equal Employment Opportunities Commission (EEOC, 2021). For physical appearance, we expand beyond the EEOC's definitions to include other features (e.g., weight) with reported bias. In each example, the negative question explicitly states a social bias that targets a member of that category. For example, within religion, some templates test biases against Muslims (e.g., associations with terrorism, violence, or misogyny), and others test biases against Jews (e.g., associations with greed or dual loyalties).

We include two intersectional biases: gender by race/ethnicity and socio-economic status by race/ethnicity. We analyze this subset separately from the other categories, as non-target construction can not be random, as one relevant demographic feature is often shared between the target and non-target, making the contrast more similar than in the non-intersectional subset of BBQ.

### 3.2 Template Construction

Table 1 shows a template example from each of the nine bias categories in BBQ. Each template is handwritten by the authors and references a specific bias reported as negative towards or having caused harm to members of a given protected category. We take any of the following as evidence or documentation of harm: research articles documenting the effects of stereotypes and biases, news articles discussing the effects of specific stereotypes or biases, Wikipedia pages listing stereotypes and biases negatively associated with a given group, and personal blogs giving first-hand accounts of harm caused by stereotypes and biases. For the two intersectional categories, we create templates with three comparisons such that one named entity is the bias target, and the second entity differs along one of the two dimensions targeted or both. For example, if the bias target is Black men, the comparisons may be between a Black man and (a) a Black woman, (b) an Asian man, and (c) an Asian woman.

**Questions** Examples come in clusters of four, with *negative* and *non-negative* question variants, with and without a disambiguating context. A negative question is one in which a social value in the US is violated *and* the bias it reflects is harmful to certain groups. A non-negative question is the complement of the negative question; it does not imply that it is not harmful. This allows us to assess whether the response rate for a given label is biased, as some labels may have a higher overall probability in the model's training data, which may affect the probability of selecting it as a response. If we only used negative questions, we would not be able to tell whether an answer was chosen due to a *question-agnostic* preference for the disadvantaged group (itself a kind of bias, but not the one we are measuring) or the specific biased association that the question is meant to draw out.

Some questions include words that indicate probabilistic judgments, such as "probably" or "likely." We include these to allow for a degree of pragmatic inference, though in some cases the biases that we test correspond to real-world differences (e.g., older adults really are more likely, on average, to have trouble hearing compared to younger adults). However, even in these cases, the attribution of a specific property to an individual in the text would not be warranted in the ambiguous contexts, as verified during our example validation (§4).

**Ambiguous Contexts** The ambiguous contexts serve to (i) introduce the general setting and the two group categories that the questions reference, and

(ii) evaluate model behavior in cases where there is insufficient evidence from the context to provide an answer. The correct answer in all the ambiguous contexts is the UNKNOWN answer option.

**Disambiguated Contexts** The disambiguated contexts provide information about which of the individuals mentioned in the ambiguous context is the answer to the negative/non-negative question. The correct answer in the disambiguated contexts is always one of the two individuals, with one individual being the answer to the negative question and the other the answer to the non-negative question.

**Balancing Templates** Li et al. (2020) note that the order in which the target and non-target phrases appear affects model behavior, and that this effect needs to be separated from measures of bias. Therefore, we permute the order of the target and non-target words for each example such that if one example from a template references 'a man in a wheelchair and a man who is walking,' another example from that same template will reference 'a man who is walking and a man in a wheelchair.'

**Number of Templates per Category** Each of the nine top-level categories and the two intersectional categories include 25 unique templates, and the race and gender categories each contain an additional 25 templates constructed using proper names. The number of examples generated for each template varies between 8 examples[2] and 200, with most templates generating at least 100 examples. In most cases, we have at least two unique templates that capture each bias (e.g., two separate templates in religion refer to the same stereotype associating Jews with greediness) in order to minimize idiosyncratic effects of a particular phrasing of the context.

### 3.3 Vocabulary

**Explicit Labels of the Bias Target** The vocabulary for the target and non-target labels is specific to each category. In the case of nationality, race/ethnicity, religion, sexual orientation, and gender, the templates are typically created from a vocabulary of group labels (e.g., "Muslim", "Buddhist", and "Jewish" are three of the 11 labels used in religion). For age, disability status, physical appearance, and socio-economic status, the labels often use a custom set of words or phrases written for each template. This customization is necessary because there are many ways to indicate that two people differ, and these descriptions or category labels differ in their appropriateness and grammatical acceptability in a given context. For example, in age, templates can reference ages (e.g., "72-year-old"), generations (e.g., "millennial"), family terms (e.g., "grandfather"), or use adjectives (e.g., "very young"). Detailed discussion of considerations in creating these labels is in Appendix A.

**Proper Names** Within gender and race/ethnicity categories, we include templates using proper names that are stereotyped of a given category (e.g., "Jermaine Washington" for a Black man, "Donna Schneider" for a White woman). Within gender, we use first names from the 1990 US census,[3] taking the top 20 most common names for people who identified themselves as male or female. Within race/ethnicity, we rely on data from a variety of sources (details in Appendix B) and always include both a given name and a family name, as both can be indicative of racial or ethnic identity in the US.

We add the strong caveat that while names are a very common way that race and gender are signaled in text, they are a highly imperfect proxy. We analyze templates that use proper names separately from the templates that use explicit category labels. However, as our proper name vocabulary reflects the most extreme distributional differences in name-ethnicity and name-gender relations, this subset still allows us to infer that if the model shows bias against some names that correlate with a given protected category, then this bias will disproportionately affect members of that category.

## 4 Validation

We validate examples from each template on Amazon Mechanical Turk. One item from each of the template's four conditions is randomly sampled from the constructed dataset and presented to annotators as a multiple-choice task. Each item is rated by five annotators, and we set a threshold of 4/5 annotators agreeing with our gold label for inclusion in the final dataset. If any of the items from a template fall below threshold, that template is edited and all four associated items are re-validated until it passes. Additional details on the validation procedure are in Appendix D. To estimate human accuracy on BBQ, we repeat the validation procedure

[2]This lower end occurs in the gender category for examples where only "man" and "woman" are slotted in.

[3]The most recent census for which this information was available (United States Census Bureau, 1990).

with a random sample of 300 examples from the final dataset. We estimate that raw human (crowd-worker annotator) accuracy on BBQ is 95.7%, and aggregate human accuracy calculated via majority vote is 99.7%. Agreement between raters is high, with a Krippendorf's $\alpha$ of 0.883.

## 5 Evaluation

**Models** We test UnifiedQA's 11B parameter model (Khashabi et al., 2020), as it achieves state-of-the-art performance on many datasets. UnifiedQA is trained on eight datasets and accepts multiple input string formats, so we include results for inputs with RACE-style (Lai et al., 2017) and ARC-style (Clark et al., 2018) formats. UnifiedQA outputs strings, so we score by exact match between the top output and each answer option.[4]

For comparison with other language models that show high performance on multiple-choice QA datasets, we also test RoBERTa (Liu et al., 2019) and DeBERTaV3 (He et al., 2021). We test both the Base and Large models to compare performance on BBQ at different model sizes. In order to test these models on BBQ, we fine-tune them on RACE (Lai et al., 2017), a multiple choice question-answering dataset, for 3 epochs with a learning rate of 1e-5 and a batch size of 16.

**Accuracy** We compute accuracy in each category and context. Within the disambiguated contexts, we further separate accuracy by whether the correct answer for the example reinforces or goes against an existing social bias in order to assess whether model performance is affected by whether a social bias is useful in answering the question.

**Bias Score** Because accuracy alone fails to capture response patterns within inaccurate answers, we introduce a bias score to quantify the degree to which a model *systematically* answers questions in a biased way. We calculate bias scores separately for the ambiguous and disambiguated contexts, as these two contexts represent model behavior in very different scenarios and require different scaling. The bias score reflects the percent of non-UNKNOWN outputs that align with a social bias. A bias score of 0% indicates that no model bias has been measured, while 100% indicates that all answers align with the targeted social bias, and -100% indicates that all answers go against the bias. Answers contribute to a positive bias score when the model outputs the bias target in the negative context (e.g. answering "the girl" for *who is bad at math?*) or the non-target in the non-negative context (e.g., answering "the boy" for *who is good at math?*). The bias score in disambiguated contexts ($s_{\text{DIS}}$) is calculated as shown below, with $n$ representing the number of examples that fall into each response group, so $n_{\text{biased\_ans}}$ represents the number of model outputs that reflect the targeted social bias (i.e., the bias target in negative contexts and the non-target in non-negative contexts), and $n_{\text{non-UNKNOWN\_outputs}}$ is the total number of model outputs that are not UNKNOWN (i.e., all target and non-target outputs).

**Bias score in disambiguated contexts:**

$$s_{\text{DIS}} = 2\left(\frac{n_{\text{biased\_ans}}}{n_{\text{non-UNKNOWN\_outputs}}}\right) - 1$$

**Bias score in ambiguous contexts:**

$$s_{\text{AMB}} = (1 - \text{accuracy})s_{\text{DIS}}$$

We scale bias scores in ambiguous contexts by accuracy to reflect that a biased answer is more harmful if it happens more often. This scaling is not necessary in disambiguated contexts, as the bias score is not computed solely on incorrect answers.[5] Although accuracy and bias score are related, as perfect accuracy leads to a bias score of zero, they reflect different model behaviors. Categories can have identical accuracies but different bias scores due to different patterns of incorrect answers.

## 6 Results

**Accuracy** Overall accuracy on BBQ is highest for UnifiedQA with a RACE-style input format at 77.8% and lowest for RoBERTa-Base at 61.4% (chance is 33.3%). However, models are generally much more accurate in the disambiguated contexts than in the ambiguous contexts (see Figure 5 in the Appendix), showing that when a correct answer is in the context, models are fairly successful at selecting it, even when that answer goes against known social biases. However, accuracy in disambiguated contexts where the correct answer aligns with a social bias is still higher than examples in which

[4] We adjust for non-content-related issues like punctuation and spelling variations. If the output matches none of the answer options after adjustment, we exclude it from analysis (3 examples excluded, or 0.005% of the data).

[5] If we scaled by accuracy in disambiguated contexts, a model that always produces biased answers would get a score of 50 because that answer is correct half the time, but the same model behavior in ambiguous contexts leads to a score of 100.